Track 2 – The Chemical Lock: Total Internal Reflection

➢ Objective

The objective of this project is to use the phenomenon of Total Internal Reflection (TIR) to identify an unknown liquid by measuring its critical angle. When light passes from a medium with a higher refractive index to a lower one, it reaches a specific angle where refraction stops and total reflection occurs. This angle is directly related to the refractive index of the liquid.

By applying Snell's Law, the refractive index can be calculated and compared with known values to determine the nature of the liquid. This shows a practical application of optics in refractometry for material analysis and identification.

➢ Mathematical Model

When light travels from a medium with higher refractive index n_1 to a medium with lower refractive index n_2, Total Internal Reflection occurs when the incident angle exceeds the critical angle θc.

$$n1 \sin\theta 1 = n2 \sin\theta 2$$

At the critical angle, the refracted ray travels along the boundary ($\theta_2 = 90°$):

$$n_1 \sin(\theta c) = n_2 \sin(90°)$$

Since $\sin(90°) = 1$:

$$\sin(\theta c) = n_2/n_1$$

Therefore:

$\theta c = \arcsin(n_2/n_1)$

Or to find the unknown refractive index:

$n_2 = n_1 \times \sin(\theta c)$

➢ Vectorization Using Dot Product

For the critical angle calculation:

$$\cos(\theta c) = \sqrt{(1 - \sin^2(\theta c))}$$

$$\cos(\theta c) = \sqrt{(1 - (n_2/n_1)^2)}$$

➢ Test Case Validation

We tested the model using:

- $n_1 = 1.60$ (glass half-cylinder)
- $\theta c = 56°$ (measured critical angle)

Calculation:

$n_2 = n_1 \times \sin(\theta c) \rightarrow n_2 = 1.60 \times \sin(56°) \rightarrow n_2 \approx 1.33$

Result: The unknown liquid is Water (n = 1.33)

➢ Implementation

The algorithm follows these steps

1. Input: n_1 (known medium), θc (measured critical angle)
2. Convert θc from degrees to radians
3. Compute: $n_2 = n_1 \times \sin(\theta c)$
4. Compare n_2 with reference values:
   - Water: n = 1.33
   - Ethanol: n = 1.36
   - Glycerin: n = 1.47
   - Benzene: n = 1.50
5. Identify the liquid with closest match
6. Check if TIR is possible: $n_1 > n_2$

The algorithm has constant complexity O(1), since it performs a fixed number of mathematical operations.

➢ Visualization

The simulation displays a glass half-cylinder with a liquid layer on its flat surface, illustrating light behavior at the glass–liquid interface under critical angle conditions.

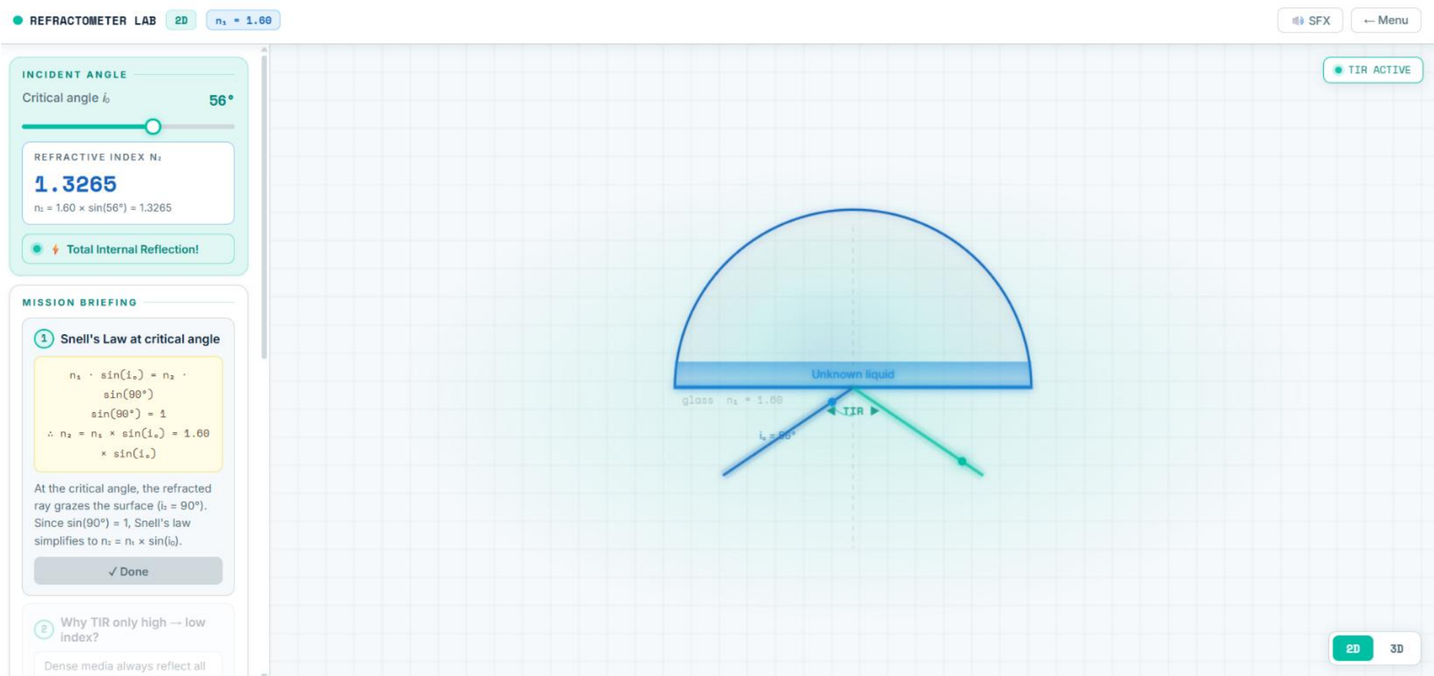
- Blue ray: Incident ray (entering from below through the curved glass surface).
- Green ray: Reflected ray inside the glass when Total Internal Reflection (TIR) occurs.

- Dashed vertical line: Normal vector at the flat interface.Light blue fill: Glass medium (n_1 = 1.60)
- Light blue region: Glass medium *(n_1 = 1.60)*.
- Upper blue layer: Liquid layer (identified in the output as Water, *$n_2 \approx 1.33$*).
- Blue curved boundary: Outer surface of the glass half-cylinder.
- Horizontal flat boundary: Interface between glass and liquid.

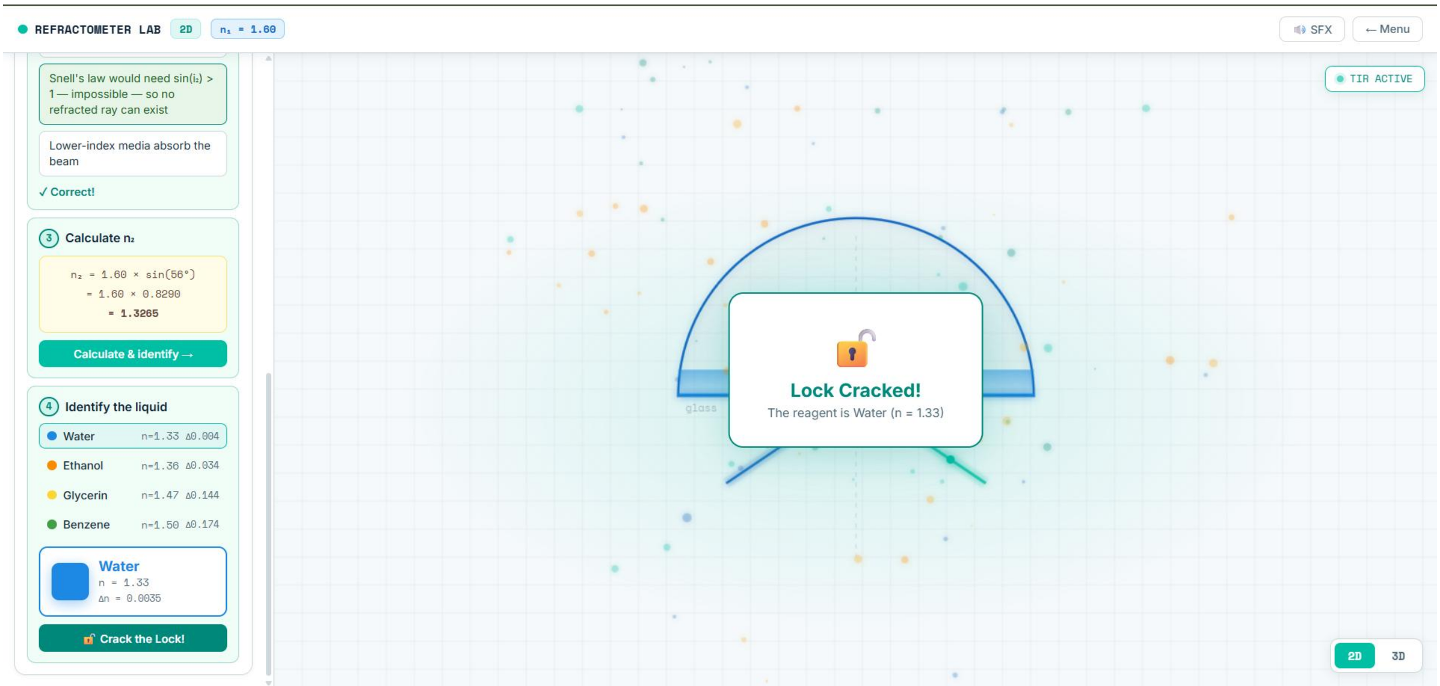
Physical Behavior:
When the incident angle reaches the critical angle ($\theta = \theta c = 56°$), the refracted ray travels exactly along the interface ($\theta_2 = 90°$). This means the light is at the boundary between refraction and total internal reflection.

The refractive index of the unknown liquid is $n_2 \approx 1.33$, meaning the medium is less dense than glass (n_1 = 1.60), making total internal reflection (TIR) possible. For $\theta < \theta c$, part of the light is transmitted into the liquid while the rest is reflected inside the glass; at $\theta = \theta c$, the refracted ray is parallel to the interface; and for $\theta > \theta c$, all the light is reflected back into the glass with no refraction.

This phenomenon is called **Total Internal Reflection (TIR).**



After the “Lock Cracked” confirmation in the simulation, the liquid is identified as water ($n = 1.33$), which validates the theoretical result and confirms the physical behavior of light at the interface.



The coordinate grid (X and Y axes) is shown in the background to provide spatial reference for angle measurement.

➢ Discussion

The simulation correctly implements Total Internal Reflection physics.

When light travels from glass ($n_1 = 1.60$) to water ($n_2 = 1.33$):

- For $\theta < 56°$: Partial refraction and reflection occur
- For $\theta = 56°$: Refracted ray travels along the boundary
- For $\theta > 56°$: Complete internal reflection occurs

The critical angle measurement provides a reliable method for identifying unknown liquids based on their refractive indices.

This principle is widely used in:

- Refractometers for chemical analysis
- Fiber optic communications for light guidance
- Gemology for identifying precious stones
- Medical diagnostics for blood analysis

Small numerical errors may appear due to floating-point precision.

The vector formulation is efficient and suitable for real-time educational simulations.

Snell's Law Visualization Schema